

venu dammalapati

Fairfax, VA

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Education

George Mason University

Aug. 2023 – May 2025

Masters of Science in Computer Science (GPA: 3.70/4)

Fairfax, Virginia

Experience

Intersect Healthcare Systems Inc.

Aug 2024 – Present

Machine Learning Engineer

Remote

- Developed and maintained Python-based backend services for healthcare analytics and ML model deployment, following secure and scalable software engineering practices.
- Built RESTful APIs for real-time data exchange between AI modules and client dashboards, ensuring reliability, low latency, and system interoperability.
- Containerized applications using Docker and orchestrated deployments with Kubernetes, collaborating with DevOps teams to support scalable cloud infrastructure and CI/CD workflows.

George Mason University

Aug 2023 – July 2024

Graduate Research Assistant

Fairfax, VA

- Designed Python-based research prototypes for AI-driven education tools, focusing on automation, data processing, and modular code architecture.
- Implemented API integrations and data pipelines to enable communication between distributed systems and experimental platforms.
- Utilized Docker for reproducible testing environments and performed basic network configuration and troubleshooting for distributed research experiments.

IBM

July 2022 – July 2023

Associate Systems Engineer

Hyderabad, India

- Contributed across the full Software Development Life Cycle (SDLC), including development, testing, deployment, and production support of enterprise applications.
- Developed scalable software components using Java, Python, and Node.js, improving application performance, reliability, and maintainability.
- Performed root-cause analysis for production issues and collaborated with cross-functional teams to implement architect-guided fixes and enhancements.

Eduskills

Jan 2022 – June 2022

AI/ML Engineer

Remote

- Built ML pipelines on AWS (Amazon SageMaker), improving training efficiency and deployment reliability.
- Developed computer vision and NLP models, improving performance by 20% and enabling scalable cloud inference.

Projects

Food Safety Incident Analysis using NLP | *Python, Streamlit, SQL, FAISS, REST APIs*

- Built an NLP-driven analytics platform using a SQL-RAG hybrid approach with FAISS, achieving 90% retrieval accuracy across structured and unstructured food safety data.
- Developed a Streamlit-based interactive dashboard and deployed containerized services using Docker and Kubernetes, enabling scalable multi-user access in a cloud environment.

News Analysis Research Tool | *Python, LangChain, NLP, FastAPI, MySQL, AWS*

- Engineered an AI-powered news summarization and sentiment analysis system using LangChain, FastAPI, and NLP techniques, reducing manual research time by 40%.
- Designed Python-SQL data pipelines and deployed microservices on AWS using Docker and Kubernetes, enabling scalable processing and improving system reliability.

Technical Skills

Languages & Frameworks: Python, Java, C, JavaScript/TypeScript, HTML/CSS, Flask, Django, FastAPI, REST APIs
Cloud/DevOps & Tools: AWS (EC2, S3), Docker, Kubernetes, GitHub Actions, CI/CD, Git/GitHub, VS Code, Eclipse, Android Studio, GCP
Data/AI & Systems: SQL/MySQL, FAISS, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch (Basics), Linux, TCP/IP, Load Balancing, VPN